

FOG (Future of Gaming)

Round 1 – Unity Developer Assignment

Round 1 – Online assessment for Unity Developer role.

1. Please **copy this document**
2. replace your **name in the document title**,
3. fill in the candidate details below, answer all questions, and submit your completed document via the provided Google Form - <https://forms.gle/TXezZ38W1LQenMNZ9>

{Candidate Name}

Instructions –

- The assessment should be done solely by yourself.
- Submit the response to assessment within 3 days.

Tips to succeed –

- Pay attention to detail
- Think user first
- Be creative
- Use AI wherever needed (Chat GPT, Midjourney, Claude, v0, etc)

All the best and don't forget to have fun!

Game Overview

Title: Escape The Lava

Core Gameplay: The player is presented with a 16x8 (rows: 16, cols: 8) grid consisting of Red Lava, Blue Diamonds, and Green Islands. The goal is to interact with the grid to collect all the diamonds before time runs out, while strictly avoiding the lava.

Technical & Gameplay Requirements

Grid & Level Design

- **Grid Size:** 16 columns x 8 rows.
- **Tile Types:**
 -  **Blue Diamonds:** Collectibles.
 -  **Red Lava:** Danger zones.
 -  **Green Islands:** Safe zones.

Interactions & Mechanics

- **Time Limit:** 30 seconds per round.
- **Lives:** The player starts with 5 lives.
- **Collecting Diamonds:** When the player clicks/taps on a Blue Diamond, it is collected and added to the score.
- **Hitting Lava:** When the player clicks/taps on Red Lava, they lose 1 life.
- **Safe Zones:** Green Islands are purely safe visual tiles (tapping them does nothing).
- **Win/Loss Conditions:**
 - **Win:** Collect all blue diamonds on the board before the timer hits 0.
 - **Game Over:** The timer reaches 0, OR the player loses all 5 lives.

Animations & Visual Feedback

Visual polish is a key part of this assessment. You must implement the following:

- **Idle Animations:** Continuous ambient animations for the Lava (bubbling/glowing) and Diamonds (floating/shining).
- **Score Animation:** A floating text/score popup animation at the exact position where a player clicks a diamond.
- **Damage Animation:** A splash or burn effect when the player taps the Red Lava.
- **Game Over Sequence:** A dedicated UI animation/screen that appears when the game is lost or won.

UI (User Interface)

- Display a countdown **Timer** (30s).
- Display the current **Lives** remaining (e.g., 5 heart icons).
- Display the current **Score / Diamonds Collected**.



Submission Format

Please submit your completed assignment using the following formats:

1. **GitHub URL:** Provide a link to your public repository containing the full Unity project. Ensure your code is clean, commented, and easy to navigate.
2. **Output Video Link:** Provide a screen recording (YouTube unlisted, Google Drive, or Loom) demonstrating a full playthrough. The video should showcase all animations, a win scenario, and a loss scenario.

Important Note: The mechanics and themes outlined below are provided strictly for gameplay context. We want you to showcase your unique creativity. Your primary focus should be designing an intuitive, easy-to-understand gameplay experience paired with **next-level animations** and visual polish.

References

1. https://youtu.be/0QVa8rEQOoM?si=NW4_BFt1yHRP8qcb
2. https://youtu.be/KpA73xWZl70?si=n6axyuG0upx-MU_x
3.  gameplay

